Rare Earth Separation Modeling with Differentiable Flowsheets

Jorge Urruela
M.S. AI Engineering-Chemical Engineering Student
The Kitchin Research Group

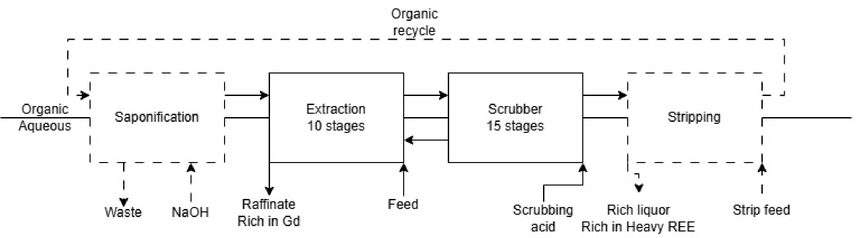
Project Motivation

- Existing flowsheet modeling options are not differentiable
- Differentiability is crucial for optimization
- Mathematical gradients are computationally expensive and/or imprecise
- REEs have similar physicochemical properties, but different functionalities
- REE processes are notably sensitive to certain factors

JAX-based modules for flowsheet modeling enable automatic differentiability, scaling from simple calculations to unit operations and all the way towards entire processes, while maintaining high performance.

The process (Lyon, et al., 2016)

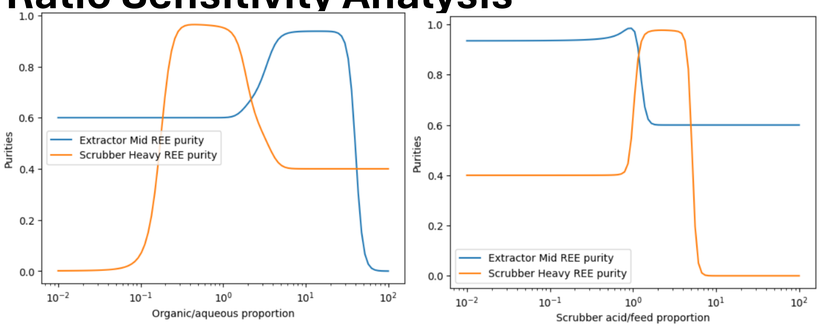
Extractor-Scrubber system to separate mid weight gadolinium (Gd) from heavy yttrium (Y) and Holmium (Ho), using PC88A and kerosene in the organic phase.



4 degrees of freedom:

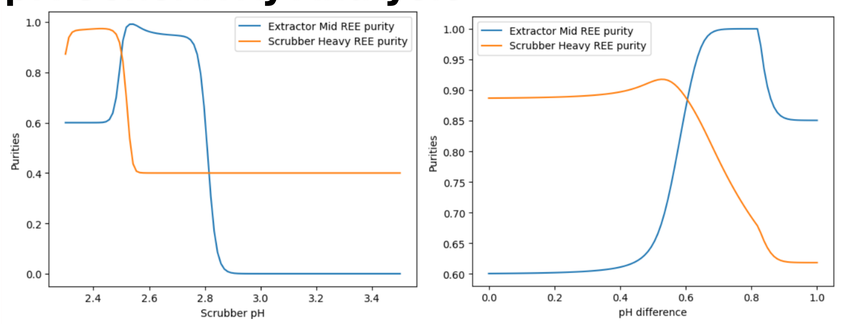
- scrubbing pH
- pH difference between extraction and scrubber
- organic phase/feed ratio
- scrubbing acid/feed ratio

Ratio Sensitivity Analysis



- Limited operation range
- Improving one purity worsens the other
  - Optimization problem
- Shape is constant for a significant range, and non constant for a limited range
- Same case for pH sensitivity

pH Sensitivity Analysis



Gradients and Optimization Problem

4 independent variables:

$$pH_{ext}, \Delta pH, r_o\left(\frac{organic}{feed}ratio\right), r_{sa}\left(\frac{scrubbing\ acid}{feed}ratio\right)$$

2 dependent variables:

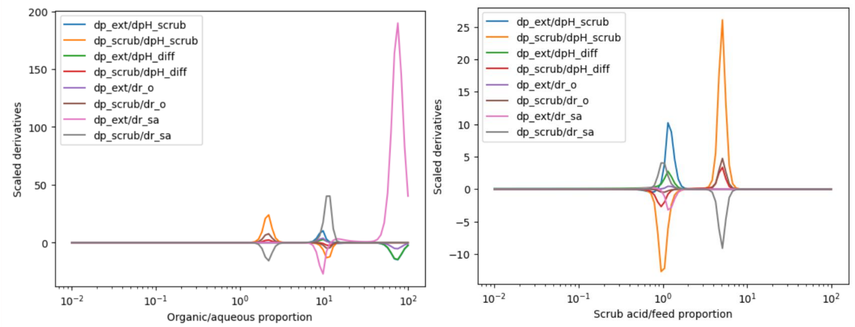
$$f(pH_{ext}, \Delta pH, r_o, r_{sa}) = p_{raff,MREE}$$
$$g(pH_{ext}, \Delta pH, r_o, r_{sa}) = p_{liq,HREE}$$

This leaves 8 gradients of interest

Non-differentiable optimization variable: minimal purity

$$\min(p_{raff,MREE}, p_{liq,HREE})$$

Normalized gradient ratio sensitivity analysis



Optimal formulation

Independent variables

- $pH_{ext} = 2.50102447$
- $\Delta pH = 0.60006841$
- $r_o = 9.76976233$
- $r_{sa} = 1.15001887$

Dependent variables:

- $p_{raff,MREE} = 0.8815124960534813$
- $p_{liq,HREE} = 0.8815124958900022$

Challenges

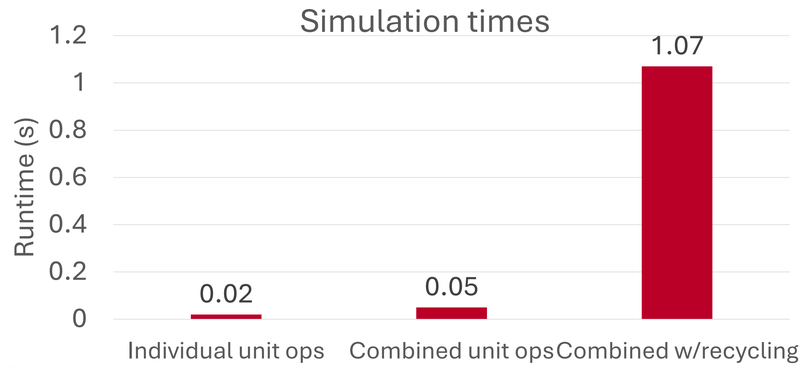
- Too many conditions are not specified by the paper, complicating the replication of the problem with the data
- Most solvers get settled at 'local' solutions where the values are constant. Bounding to avoid these is important
- Taking advantage of differentiability to solve the optimization problem
- Recycling does not converge at a constant number of iterations. In fact, it varies greatly as the variables change

Insights

- Optimistix recycling scales up computational power required, but maintains differentiability
- Typical solvers that are gradient-based are very likely to settle at a local maximum, hence unable to find the real optimal solution
  - Controlling operation ranges is key
- Temperature gradients are not as important, but working at a low but valid temperature is
- Sensitivity analysis for the derivatives have an unusual shape, but provide valuable insights on the ranges of operation

Next steps

- Modify temperatures to replicate the results accurately
- Validate unit operations individually
- Obtain and compare mathematical gradients ($\Delta y/\Delta x$), second gradients
- Better bounding to solve the optimization problem while maintaining differentiability
- Propose a surrogate model: useful for larger, more complex processes



References

Lyon, K. *et al.* (2016) Enhanced Separation of Rare Earth Elements. *International Mineral Processing Congress*. Idaho National Laboratory.